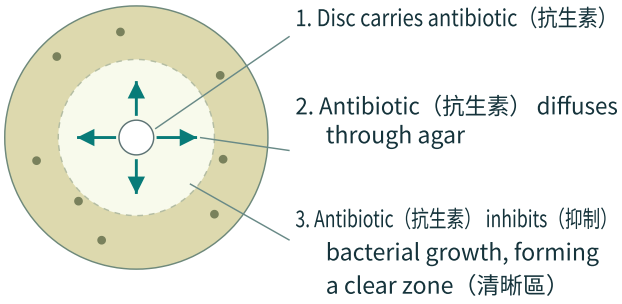
Name: _____ Class: _____ No.: _____

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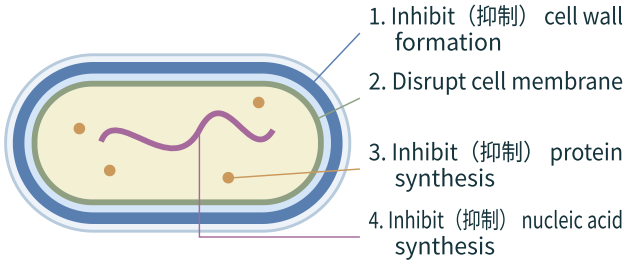
Learning points

A. Experimental principle: antibiotic（抗生素） diffusion and clear-zone（清晰區） formation



Schematic, not to scale; it does not prove complete killing.

B. Structures and possible action sites



The pale outer layer represents a capsule (not present in all bacteria); blue is the cell wall, green is the cell membrane, and the purple strand is DNA. These mechanisms are not proved by this disc experiment.

Uses of antibiotics（抗生素）

- Antibiotics（抗生素） mainly treat bacterial infections; they cannot treat viral colds or influenza.

How clear zones（清晰區） form

- Antibiotic（抗生素） diffuses outward from the disc, and its concentration generally decreases with distance. A clear zone（清晰區） forms only where the concentration remains sufficient to inhibit（抑制） this bacterium. Greater susceptibility（對抗生素的敏感性） (growth inhibited（抑制） at a lower antibiotic（抗生素） concentration) allows a larger clear zone（清晰區） to form.
- A clear zone（清晰區） indicates no visible bacterial growth at that location, demonstrating that the antibiotic（抗生素） sample can inhibit（抑制） bacterial growth; it does not prove that all bacteria in the zone have died.

Antibiotic resistance（抗生素抗性／抗藥性） and selection

- Why does the original antibiotic（抗生素） in the scenario lose effectiveness? Antibiotics（抗生素） make survival harder for susceptible（對抗生素敏感的） bacteria, while resistant（具抗藥性的） bacteria are more likely to survive, reproduce（繁殖） and pass on resistance traits（抗藥性特徵）. As resistant（具抗藥性的） bacteria increase in proportion and occupy the whole population, the original antibiotic（抗生素） becomes ineffective.
- Antibiotic（抗生素） overuse（過度使用） or misuse（不當使用） adds unnecessary selection pressure（選擇壓力）, promoting the survival and reproduction（繁殖） of resistant（具抗藥性的） bacteria and potentially making infections harder to treat.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

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Learning reflection

Your experimental results:

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.

Revisions, if any (keep the first response readable; this is not an additional question)